Publishing 'AI-Driven Design for SIW Filters'

Venue landscape, AI-disclosure requirements, and a recommendation

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What this paper actually is

Two different papers could be written from this project, and they point at different venues:

1. **An SIW filter synthesis paper.** Chebyshev coupling-matrix synthesis, calibrate-and-invert (measure k/Qe on a small closed structure, invert the textbook mapping, assemble and confirm), applied to a dual-mode and an inline-cascade SIW cavity filter, with a fabricated board. This is a normal RF/microwave paper that happens to have been designed with heavy AI assistance.
2. **An agentic-design-automation paper.** An LLM coding agent drove an open-source FEM solver (Palace) through a design loop end to end — wrote the meshing/optimization code, ran the EM solves, generated the fabrication outputs, and in the process produced a documented taxonomy of **fifteen-plus silent modelling bugs**, none of which raised an error. The filter is the case study; the finding is about the process.

These have different homes. Framing (1) fits classic RF/microwave venues. Framing (2) fits the newer, smaller world of venues explicitly about agentic AI and LLM-driven engineering — and that world does **not** currently include the RF/microwave-specific journals, whose "AI for microwave filters" scope almost always means *trained ML surrogate models* (neural networks predicting S-parameters or synthesizing geometry), not an LLM agent writing and iterating the design code itself. Picking the wrong framing for a venue is the single biggest fit risk here.

Do these venues even accept AI-assisted work?

Yes — every venue considered below permits it. None forbid AI assistance; all require **disclosure**, not permission-seeking, and none allow an AI system to be listed as an author.

Publisher	Authorship	Disclosure required	Where
IEEE (LAD, IMS, IMFW, JSTEAP)	AI cannot be an author	Yes, in Acknowledgments: name the tool, name the sections, briefly explain how it was used. Grammar/editing-only use is exempt.	IEEE generative-AI policy, IEEE RAS guidelines
Wiley (IJRFMiCAE)	AI cannot be an author	Yes, full disclosure in a dedicated appendix; author takes full responsibility for the content.	Wiley generative-AI author guidelines
MDPI (Electronics)	AI cannot be an author	Yes, disclosure statement in Acknowledgments naming the tool, plus a Materials-and-Methods description of specifically how it was used. A vague "ChatGPT improved this manuscript" does not meet their standard.	MDPI generative-AI authorship policy

Separately and unconditionally: using AI to fabricate or manipulate data, figures, or results is a research-integrity violation, independent of the disclosure question. Nothing here is at risk on that front — every number in this project traces to a real Palace solve or a real gerber/drill file, and the whole point of the bug taxonomy is that errors were *found and fixed*, not hidden.

Practical takeaway for this specific paper: because the paper's subject is AI-driven design, the disclosure isn't a compliance footnote — it's the paper's own methods section. Be specific about what the agent did (wrote the meshing/optimization/export code, ran the EM solves, found and fixed the bugs described in the results, drafted portions of the text) rather than a blanket statement. That specificity is exactly what MDPI and IEEE both say separates an adequate disclosure from an inadequate one.

Venue options, by framing

Framing 1 — RF/microwave synthesis paper

Venue	Type	Fit	Deadline (from today, 2026-09-04)
Int'l J. of RF and Microwave CAE (Wiley)	Journal, rolling	Exactly on-topic (CAD/EDA for RF/microwave); no deadline pressure	rolling
MDPI Electronics	Journal, rolling, open access	Broad enough, has run ML-for-microwave-filter themes before	rolling
IEEE MTT-S IMS 2027 (San Antonio, May 2027)	Flagship conference	Best prestige match; this cycle explicitly expanded scope for AI/ML-driven RF design	**Jan 6, 2027** — reachable
IEEE JSTEAP "AI for EM/Antennas/Microwave Circuits" special issue	Journal special issue	Ideal scope on paper	this cycle's deadline (Dec 15, 2025) already passed; watch for the next call
IEEE MTT-S IMFW (Filter Workshop, 3-page format)	Workshop	Filter-specific audience, short-paper format suits a methods/negative-result piece	this cycle's (Oct 1, 2025) passed; annual, next around Oct 2026 for IMFW 2027
EuMC / EuMW (London, Oct 2026)	Flagship conference	Good fit	this cycle's deadline (Mar 27, 2026) already passed

Framing 2 — agentic-AI / LLM-driven design-automation paper

Venue	Type	Fit	Deadline (from today)
IEEE LAD ("LLM-Aided Design", `iclad.ai`, IEEE CEDA)	Conference, IEEEXplore	**Best topical match found.** LAD 2026's own stated theme was "agentic optimization and scaling inference-time methods" — a near-exact description of the calibrate-and-invert + local-GP loop this project ran.	LAD 2026 already ran (deadline Mar 2, 2026; conference Jul 30–31, 2026, Stanford). **LAD 2027 not yet posted** — watch `iclad.ai`; historical pattern suggests a spring 2027 deadline.

Venue	Type	Fit	Deadline (from today)
arXiv (`eess.SY` or `cs.AI`)	Preprint, no review	Where the closest comparable paper currently lives (see below)	immediate, no deadline
IEEE Access	Journal, rolling, open access	Broad enough to accept an agentic-AI-methods paper even without a novel trained model; fast review	rolling
IEEE CAI ("Agents for Science and Engineering" session)	Conference special session	Direct topical fit	CAI 2026 already ran (May 2026, Granada); watch for CAI 2027

A directly comparable prior work: *From Prompt to Prototype: Towards a Frontier LLM Driven RF Engineering Workflow* (arXiv:2608.31006, Aug 2026) — an LLM agent operating CST Studio Suite, Keysight ADS, and KiCad via their scripting interfaces to design a GNSS antenna system (patch, SAW prefilter, two-stage LNA) end to end in about two hours. Same category of contribution as this project's, currently only on arXiv with no venue attached — a sign this specific niche (agentic LLM + real EM/CAD tools, not a trained surrogate) does not yet have a single settled home. Worth citing as related work either way.

Recommendation

If the point of the paper is genuinely "an AI coding agent drove an open-source solver through a design loop and the process itself is the finding" (framing 2 — the approach actually followed: Palace, not CST, as the solver of record; AI writing and iterating code, not a trained surrogate model):

1. **Post to arXiv now.** Establishes priority immediately, costs nothing, and is where the one directly comparable paper already lives.
2. **Target IEEE LAD 2027** when its CFP opens (watch `iclad.ai`). This is the venue whose own stated interests (agentic optimization, inference-time scaling, LLM-driven design tools) match the actual contribution most closely of everything found.
3. **IEEE Access** is the fastest realistic peer-reviewed option if IEEE credit is wanted sooner than LAD 2027's cycle allows — it doesn't require a novel ML model, just a real methods contribution, which this has (the bug taxonomy, the calibrate-and-invert method, the fabricated result).

If instead the paper should foreground the filter-synthesis contribution and the fabricated result, with the AI-driven process as a secondary disclosure rather than the headline: **IJRFMiCAE or MDPI Electronics** now (rolling, no deadline pressure), with **IMS 2027** (Jan 6, 2027) as a stretch target for the flagship conference. Nothing prevents submitting both a framing-2 preprint/LAD paper and a framing-1 journal paper from the same underlying work, provided each stands on its own contribution and neither double-publishes the same claims as new.

Sources

- IEEE generative-AI author-disclosure policy
- IEEE RAS guidelines for generative AI usage
- MDPI's generative-AI authorship policy
- Wiley generative-AI guidelines for authors (PDF)
- IEEE LAD 2026 call for papers
- ICLAD / LAD conference series
- From Prompt to Prototype (arXiv:2608.31006)
- IEEE MTT-S IMS 2027 call for papers
- IMFW 2026 call for papers
- EuMW 2026 call for papers
- IEEE JSTEAP AI special issue call
- International Journal of RF and Microwave CAE (Wiley)
- IEEE CAI 2026

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cd docs && pandoc --pdf-engine=xelatex PUBLICATION_VENUES.md -o PUBLICATION_VENUES.pdf
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